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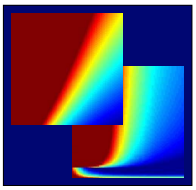
CS/CNS/EE 156a Learning Systems
Caltech - Fall 2025
https://caltech.instructure.com/courses/8882
(Learning From Data campus version)



Homework # 3

Due Monday, October 20, 2025, at 2:00 PM PDT

*Definitions and notation follow the lectures. All questions have multiple-choice answers (***[a], [b], [c], ...***). Collaboration is allowed but without discussing selected or excluded choices. Your solutions must be based on your own work. See the initial "***Course Description and Policies***" handout for important details about collaboration, open book, and chatGPT policies.*

Note about the homework

- Answer each question by deriving the answer (carries 6 points) then selecting from the multiple-choice answers (carries 4 points). You can select 1 or 2 of the multiple-choice answers for each question, but you will get 4 or 2 points, respectively, for a correct answer. See the initial "**Course Description and Policies**" handout for important details.
- The problems range from easy to difficult, and from practical to theoretical. Some problems require running a full experiment to arrive at the answer.
- The answer may not be obvious or numerically close to one of the choices, but one (and only one) choice will be correct if you follow the instructions precisely in each problem. You are encouraged to explore the problem further by experimenting with variations on these instructions, for the learning benefit.
- You are encouraged to take part in the Piazza discussion forum. Please make sure you don't discuss specific answers, or specific excluded answers, before the homework is due.

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- **Generalization Error**

1. The modified Hoeffding Inequality provides a way to characterize the generalization error with a probabilistic bound

$$\mathbb{P}\left[|E_{in}(g) - E_{out}(g)| > \epsilon\right] \leq 2Me^{-2\epsilon^2 N}$$

for any $\epsilon > 0$. If we set $\epsilon = 0.05$ and want the probability bound $2Me^{-2\epsilon^2 N}$ to be at most 0.03, what is the least number of examples N (among the given choices) needed for the case $M = 1$?

[a] 500

[b] 1000

[c] 1500

[d] 2000

[e] More examples are needed.

2 e^(−2(0.05)²N) ≤ 0.03
e^(−0.005N) ≤ 0.015
−0.005N ≤ ln(0.015)
N ≥ (ln(2/0.03)) / (2ε²)
N ≥ ln(66.6667) / 0.005
N ≥ 4.1997 / 0.005
N ≥ 839.94

2. Repeat for the case $M = 10$.

[a] 500

[b] 1000

[c] 1500

[d] 2000

[e] More examples are needed.

2·10·e^(−2(0.05)²N) ≤ 0.03
e^(−0.005N) ≤ 0.03 / 20 = 0.0015
−0.005N ≤ ln(0.0015)
N ≥ (ln(2·10/0.03)) / (2ε²)
N ≥ ln(666.6667) / 0.005
N ≥ 6.5023 / 0.005
N ≥ 1300.46

3. Repeat for the case $M = 100$.

[a] 500

[b] 1000

[c] 1500

[d] 2000

[e] More examples are needed.

2·100·e^(−2(0.05)²N) ≤ 0.03
e^(−0.005N) ≤ 0.03 / 200 = 0.00015
−0.005N ≤ ln(0.00015)
N ≥ (ln(2·100/0.03)) / (2ε²)
N ≥ ln(6666.6667) / 0.005
N ≥ 8.8049 / 0.005
N ≥ 1760.98

- **Break Point**

4. As shown in class, the (smallest) break point for the Perceptron Model in the two-dimensional case ($\mathbb{R}^2$) is 4 points. What is the smallest break point for the Perceptron Model in $\mathbb{R}^3$? (i.e., instead of the hypothesis set consisting of separating lines, it consists of separating planes.)

[a] 4

[b] 5

[c] 6

[d] 7

[e] 8

In R^d, VC dimension has breakpoint d+1.
Therefore the smallest break point is (VC dimension) + 1 = d+1 + 1 = d+2
For d=3, break pooint is 3+2=5

- **Growth Function**

5. Which of the following are possible formulas for a growth function $m_{\mathcal{H}}(N)$:

$$\begin{array}{rl} \text{i)}\ 1+N & \text{iv)}\ 2^{\lfloor N/2 \rfloor} \\ \text{ii)}\ 1+N+\binom{N}{2} & \text{v)}\ 2^N \\ \text{iii)}\ \sum_{i=1}^{\lfloor \sqrt{N} \rfloor} \binom{N}{i} & \end{array}$$

where $\lfloor u \rfloor$ is the biggest integer $\leq u$, and $\binom{M}{m} = 0$ when $m > M$.

[a] i, v

[b] i, ii, v

[c] i, iv, v

[d] i, ii, iii, v

[e] i, ii, iii, iv, v

i) 1 + N

= Σ_{i=0}^1 C(N,i) → VC dimension 1 growth -> valid.

ii) 1 + N + C(N,2)

= Σ_{i=0}^2 C(N,i) → VC dimension 2 growth -> valid.

iii) Σ_{i=1}^{⌊√N⌋} C(N,i)

VC dimension would grow with N (not fixed) and missing C(N,0)=1; not polynomial of fixed degree nor 2^N -> invalid.

iv) 2^{⌊N/2⌋}

unsure but it must be false by logic given I am sure about my other answers ;)

v) 2^N

Infinite VC dimension -> valid.

- **Fun with Intervals**

6. Consider the "2-intervals" learning model, where $h\colon \mathbb{R} \to \{-1, +1\}$ and $h(x) = +1$ if the point is within either of two arbitrarily chosen intervals and -1 otherwise. What is the (smallest) break point for this hypothesis set?

[a] 3

[b] 4

[c] 5

[d] 6

[e] 7

Each interval on number line contributes 2 degrees of freedom with both its endpoints. Thereforem in 2 intervals, we have a VC dimension of 2*2=4, making the smallest break point 4+1=5.

7. Which of the following is the growth function $m_H(N)$ for the "2-intervals" hypothesis set?

Cases:

- No intervals (both empty): C(N+1, 0) = 1
- Exactly one interval where one interval is non empty, the other is empty (or we just use one interval): choose 2 cut points -> C(N+1, 2)
- Two disjoint intervals where both intervals are non-empty and separate: choose 4 cut points -> C(N+1, 4)
- total = C(N+1, 4) + C(N+1, 2) + 1

[a] $\binom{N+1}{4}$

[b] $\binom{N+1}{2} + 1$

[c] $\binom{N+1}{4} + \binom{N+1}{2} + 1$

[d] $\binom{N+1}{4} + \binom{N+1}{3} + \binom{N+1}{2} + \binom{N+1}{1} + 1$

[e] None of the above

8. Now, consider the general case: the "M-intervals" learning model. Again $h : \mathbb{R} \to \{-1, +1\}$, where $h(x) = +1$ if the point falls inside any of M arbitrarily chosen intervals, otherwise $h(x) = -1$. What is the (smallest) break point of this hypothesis set?

[a] M

[b] $M + 1$

[c] M^2

[d] $2M + 1$

[e] $2M - 1$

As justified in (6), VC dimension = 2M (because we can shatter any set of 2M points by carefully placing our M intervals to get any labeling); break point = VC dim + 1

Therefore break point = 2M + 1

• Convex Sets: The Triangle

9. Consider the "triangle" learning model, where $h : \mathbb{R}^2 \to \{-1, +1\}$ and $h(\mathbf{x}) = +1$ if $\mathbf{x}$ lies within an arbitrarily chosen triangle in the plane and -1 otherwise. Which is the largest number of points in $\mathbb{R}^2$ (among the given choices) that can be shattered by this hypothesis set?

[a] 1

[b] 3

[c] 5

[d] 7

[e] 9

A triangle is a convex polygon w/ 3 sides.

Therefore VC dimension is 2*3+1=7.

Hence there must exists a configuration of 7 points in a plane (i.e 7 points arranged on a circle) such that for any possible labeling of these points with +-1, there exists sone triangle that perfectly achieves that labeling/

• Non-Convex Sets: Concentric Circles

10. Compute the growth function $m_{\mathcal{H}}(N)$ for the learning model made up of two concentric circles around the origin in $\mathbb{R}^2$. Specifically, $\mathcal{H}$ contains the functions which are $+1$ for

$$a^2 \leq x_1^2 + x_2^2 \leq b^2$$

and -1 otherwise, where a and b are the model parameters. The growth function is

[a] $N+1$

[b] $\binom{N+1}{2}+1$

[c] $\binom{N+1}{3}+1$

[d] $2N^2+1$

[e] None of the above

We can transform this 2D problem into 1D.

Let z = x_1^2 + x_2^2

The hypothesis becomes:

h(z) = +1 if a' ≤ z ≤ b' (note a' = a² and b' = b²)

h(z) = -1 otherwise

This is exactly the positive intervals model on a 1D line from lecture. I will walk through it.

- -1 case: We can always create a dichotomy where all points are -1 by choosing an interval that contains none of the points (i.e. a' > b', or an interval between two points where there are none). This gives 1 dichotomy.
- Cases with +1 points: Select a block of consecutive points to be +1. We can count these by the size of the block:

  Blocks of size 1: $\{z_1\}, \{z_2\}, \ldots, \{z_N\}$ (N dichotomies)

  Blocks of size 2: $\{z_1, z_2\}, \{z_2, z_3\}, \ldots, \{z_(N-1), z_N\}$ (N-1 dichotomies)

  ...

  Block of size N: $\{z_1, z_2, \ldots, z_N\}$ (1 dichotomy)

The total number of dichotomies is N + (N-1) + ... + 1, which is equal to N(N+1)/2. or C(N+1, 2)

Hence b is correct.